

EDUCATION

Bachelor of Science in Software Engineering with Minor in Mechatronics

August 2025

GPA: 3.7/4.0

- Coursework: Networks, Operating Systems, Computer Graphics, Data Structures & Algorithms, CPU Architecture

EXPERIENCE

Software Intern

September 2023 – August 2024

- Improved JOIN **performance by 20%** in a C++ database engine by caching intermediate arithmetic operations and live objects, eliminating repeated recalculations across later join and grouping stages.
- **Resolved authentication failures** under concurrent load by redesigning file I/O so that only one writer could modify credentials atomically, while readers used consistent, read-only snapshots of the authentication data file.
- Improved **system performance by 15%** by stopping columns meant only for display from being processed during sorting, merging, and result building intermediate steps, by making each step request only the data it actually needs.
- Reduced **server startup time by 8%** by reusing objects and pooling memory to avoid repeated allocations.
- Benchmarked query execution using **Visual Studio profilers**, resolving **4+ bottlenecks** in query evaluation.

Software Payload Developer

September 2024 – May 2025

Student Design Team

- Built **embedded flight firmware in C++** for a rocket payload that autonomously collected airborne microbial samples at ~30,000 ft. using live sensor data to trigger valves and pressurization at set flight conditions.
- Built custom C++ drivers for IMU, altimeter, and pressure sensors and integrated them into a real-time data pipeline using FreeRTOS queues with a **<10 ms end-to-end latency**, filtering noise to ensure reliable in-flight decisions.
- Structured the software for reliability and real-time operation, separating sensing, decision logic, and hardware control so the system remained 100% responsive during launch and flight.

Assistant Researcher

May 2022 – August 2022

- Built a distributed learning system on **Raspberry Pi** sensor devices, where each node trained AI locally on temperature and humidity data and sent compact model updates to a central service, preserving device autonomy.
- Optimized Kubernetes scheduling with CPU tuning, increasing training throughput by 12% on 8-node clusters.

PROJECTS

Static Language Compiler | (Rust) | [GitHub](#)

- Built a **full static compiler in Rust** (8K+ LOC) that detects semantic and type errors at compile time and emits native **x86_64** code via a custom backend or optionally lowering to **LLVM** for code generation.
- Implemented the complete compilation pipeline, from parsing and static typing to SSA-based IR and machine code, enabling correct optimization through control-flow graphs, dominator analysis, phi-nodes, and SSA renaming.
- Designed a linear-scan register allocator and x86-64 codegen supporting arithmetic, branching, and function calls.

Neural Network Library | (Python/Cuda) | [GitHub](#)

- Built a modular **deep learning** framework from scratch supporting dense and convolutional layers, achieving **98% accuracy on medical imaging dataset**, with optional **CUDA** acceleration providing up to 7x training speedup.
- Implemented gradient descent backpropagation with an abstract interface for activation and loss functions.

Digital Pneumatic Thermostat | (C/C++/STM32) | [GitHub](#)

- Led a 5-person team to design a **Google home** connected device to retrofit into older buildings with HVAC control.
- Developed a low-power C++ GUI on STM32 for real-time temperature and settings visualization via ePaper display.
- Implemented I2C sensor interfaces and **PID** control to regulate pneumatic output for HVAC temperature control.

TECHNICAL SKILLS

Languages: C++, Rust, Python, C, CUDA, Java, Bash

Skills: LLVM, Concurrency, Memory Allocators, Operating Systems, OpenGL, Distributed Systems, Linux, Git